Slip No.1- Arithemetic

Q6. How many general purpose registers are there in Atmega 16?

Ans. It has 32 built-in registers from R0 through R31. These registers help connect the CPU to the external peripheral devices.

Q7. Write different data types (with values) used in C-program for Atmega 16.

Ans. **Integer Data Type**

The integer datatype in C is used to store the integer numbers (any number including positive, negative and zero without decimal part). Octal values, hexadecimal values, and decimal values can also be stored in int data type in C.

- **Range:** -2,147,483,648 to 2,147,483,647
- **Size:** 4 bytes
- **Format Specifier:** %d

Character Data Type

Character data type allows its variable to store only a single character. The size of the character is **1 byte**. It is the most basic data type in C. It stores a single character and requires a single byte of memory in almost all compilers.

- **Range:** (-128 to 127) or (0 to 255)
- **Size:** 1 byte
- **Format Specifier:** %c

Float Data Type

In C programming, **float data type** is used to store single precision floating-point values. These values are decimal and exponential numbers.

- **Range:** 1.2E-38 to 3.4E+38
- **Size:** 4 bytes
- **Format Specifier:** %f

Slip No. 2 LED blinking

Q7. What are the types of LED's?

1. Smart LED
2. RGB (Red,Green,Blue) LED
3. 5mm LED

Slip No. 4 Intrusion detection (IR sensor)

Q6. Write the working principle of IR Sensor.

An IR sensor can measure the heat of an object as well as detects the motion

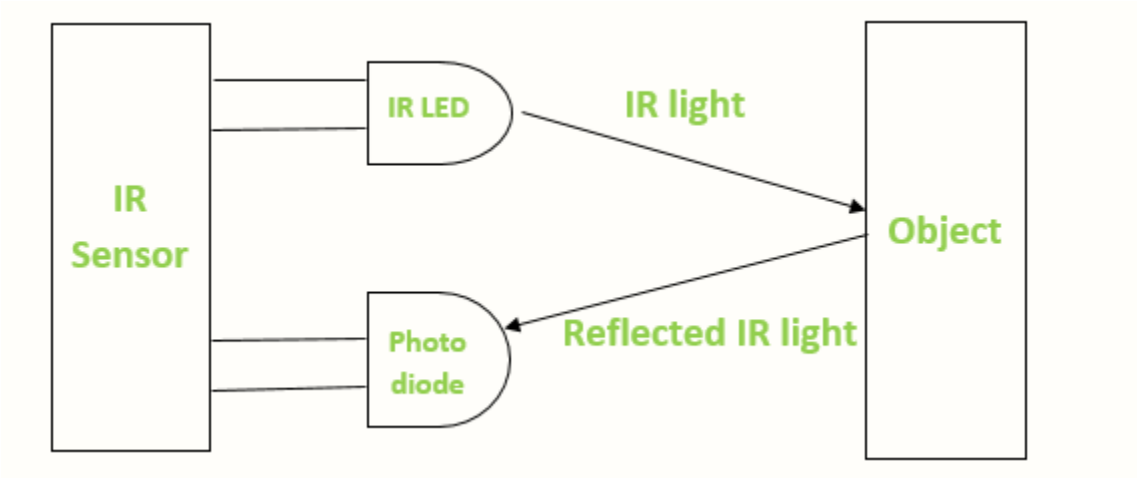
Types of IR Sensor

There are two types of IR sensors are available and they are,

- Active Infrared Sensor
- Passive Infrared Sensor

NOTE: IN PRACTICAL WE ARE USING ACTIVE INFRARED SENSOR

1. There are two diodes in the IR sensor module.
2. IR transmitter and IR receiver(photodiode).
3. The emitter is an IR LED and the detector is an IR photodiode. The IR photodiode is sensitive to the IR light emitted by an IR LED.
4. When the IR transmitter emits radiation, it reaches the object and some of the radiation reflects back to the IR receiver. Based on the intensity of the reception by the IR receiver, the output of the **sensor** defines.



Applications of IR Sensor

1. Night Vision Devices
2. IR Imaging Devices
3. TV's

Q7. How many ports are there for Atmega 16 microcontroller? List their names.

Ans. The ATmega16 microcontroller has four 8-bit ports, giving it a total of 32 general-purpose input/output (GPIO) pins.

PORTA, PORTB, PORTC & PORTD

Slip no. 9 Switch

Q6. What is a relay and explain its working.

A relay is an electrically operated switch that uses a low-power signal to control a high-power circuit, acting as an automatic on/off switch for electrical systems.

Working principle:

1. A low power electrical signal is applied to the electromagnetic coil.
2. When the current flows through the coil, magnetic field is produced.
3. The magnetic field attracts the armature.
4. The mechanical movement of armature opens or close the electrical contacts.
5. So when the contact closes circuit gets complete and the LED turns on.
6. And when the low power electrical signanl is not applied to the electromagnetic coil no magnetic field is produced and thus armature comes to its original position due to spring and the contact opens which ultimately turns off the LED.

Slip no. 10 : Bluetooth

Q6. What is the working range of Bluetooth Module (HC-05).

Ans. 10 meter (33 ft.)

Q7. Which registers (SFR's) are used while performing USART operation. List them.

Ans. The registers (SFRs) used for UART (or USART) operation in the ATmega16 are:

- UDR
- UCSRA
- UCSRB
- UCSRC
- UBRRH
- UBRRL

Slip no. 11: DAC

Q6. What is the bit size for converting digital data into analog form?

Ans. 8 bit

Q7. List the features of DAC.

Ans. Accuracy: Accuracy describes how closely the DAC's actual output voltage matches the ideal voltage for a given digital input.

Settling time: This is the time it takes for the DAC's output to stabilize within a specified accuracy range of its final value after the digital input code changes.

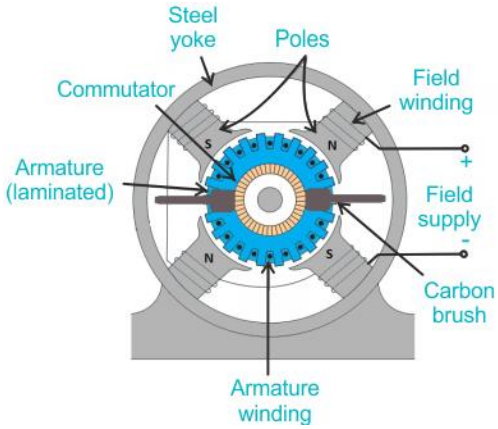
Slew Rate: The maximum speed at which the output voltage can change.

Slip no. 12 : DC motor

Explain the working principle of DC Motor.

Ans. Working Principle of DC Motor

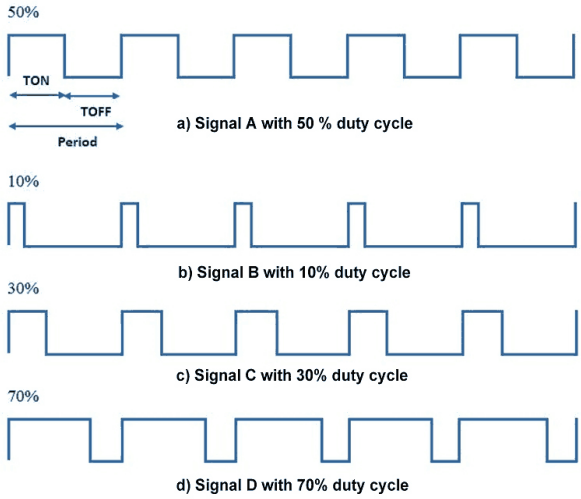
1. The basic working principle of a DC motor is that whenever a current-carrying conductor is placed in a magnetic field, it experiences a force. This force is known as the Lorentz force, and it is responsible for the rotation of the motor's armature.
2. The armature is a rotating part of the motor that is made up of coils of wire.
3. When current flows through these coils, they create a magnetic field that interacts with the magnetic field of the permanent magnets or electromagnets that are also part of the motor.
4. This interaction creates a force that causes the armature to rotate.
5. The rotation of the armature is transferred to the output shaft of the motor, which can then be used to do work, such as driving a pump, fan, or other machine.



Q7. What are PWM signals & why they used for controlling DC motor?

Ans. What is Pulse Width Modulation?

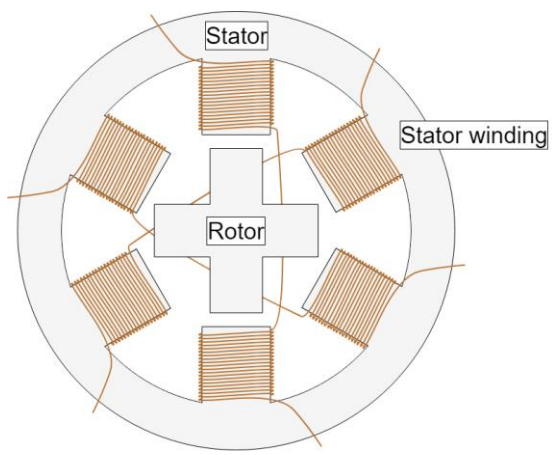
Pulse-width modulation, commonly known as PWM, is a modulation method that changes the pulse signal's width in electrical systems to regulate the average power supplied to a load.



Slip no. 13: Stepper motor

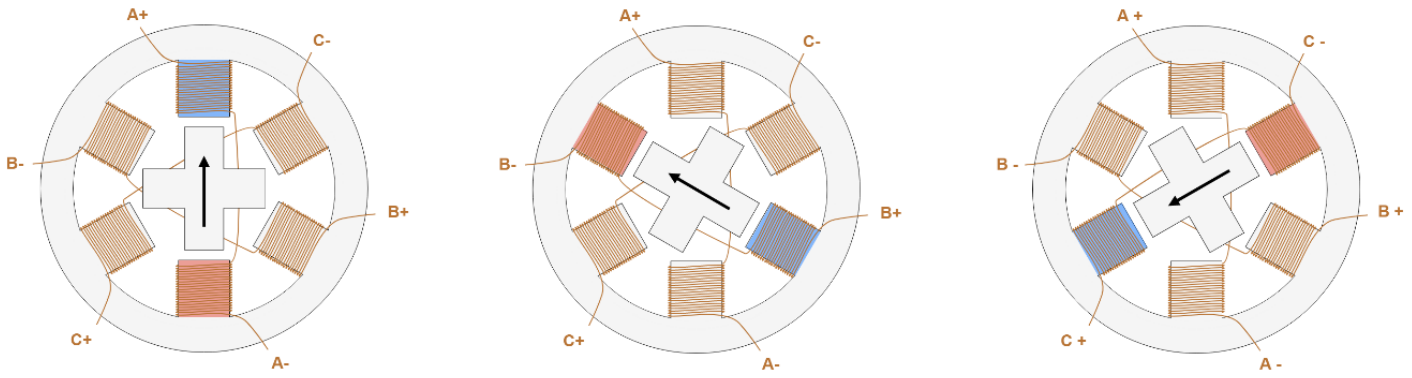
Q6. Explain working principle of Stepper Motor.

Ans. stepper motors have a stationary part (the stator) and a moving part (the rotor). On the stator, there are teeth on which coils are wired, while the rotor is either a permanent magnet or a variable reluctance iron core.



The basic working principle of the stepper motor is the following: By energizing one or more of the stator phases, a magnetic field is generated by the current flowing in the coil and the rotor aligns with this field. By supplying different phases in sequence, the rotor can be rotated by a specific amount to reach the desired final position.

When coil B is energized, the rotor rotates clockwise by 60° to align with the new magnetic field. The same happens when coil C is energized.



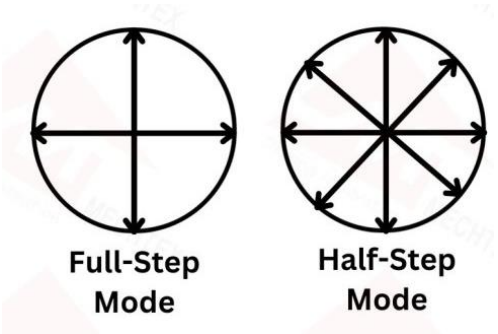
Q7. What is Full Step & Half Step mode.

- **Full Step Mode**

The full step is the basic stepping mode used to control the movement of the motor. In this mode, the motor rotates to one full step for every electrical pulse sent to the coil.

- **Half-Step Mode**

Half-step mode is a stepping mode that offers a balance between full-step and micro-stepping mode. In this mode, the motor takes half the size of the step as compared to the full-step mode steps.



Slip no. 16 : LED blinking user timer

List the registers used for Timer operations in AVR Atmega 16.

TCNTn: Timer / Counter Register

Every timer has a timer/counter register. It is zero upon reset. We can access value or write a value to this register. It counts up with each clock pulse.

TOVn: Timer Overflow Flag

Each timer has a Timer Overflow flag. When the timer overflows, this flag will get set.

TCCRn: Timer Counter Control Register

This register is used for setting the modes of timer/counter.

OCRn: Output Compare Register

The value in this register is compared with the content of the TCNTn register. When they are equal, the OCFn flag will get set.